UNIT - I

MANAGERIAL ECONOMICS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Introduction: nature, meaning, significance, functions, advantages
- Demand: concept, function, law of demand
- Demand elasticity: types and measurement
- Demand forecasting: factors governing forecasting, methods
- Managerial Economics and Financial Accounting and Management

1. Introduction to Managerial Economics

1.1 Meaning

- **Economics** is the study of how scarce resources having alternative uses are allocated to satisfy unlimited human wants.
- **Managerial Economics** = application of economic theory + decision science tools to solve business problems and take rational decisions.
- Also called **Business Economics** or **Applied Economics**. It is a bridge between economic theory and managerial practice.

Definition (Spencer & Siegelman): "Managerial Economics is the integration of economic theory with business practice for the purpose of facilitating decision making and forward planning by management."

Definition (Mansfield): "Managerial Economics is concerned with application of economic concepts and economic analysis to the problems of formulating rational managerial decisions."

1.2 Nature of Managerial Economics

- **Science and Art** : uses scientific methods and also needs skill to apply them.

- **Micro-economic in nature** : deals with the individual firm, not the whole economy.
- **Pragmatic**: practical, solves real business problems.
- **Normative** (prescriptive): tells what ought to be done, giving a goal-oriented approach.
- **Uses macro-economics also** : business environment (inflation, GDP, interest rates) affects the firm.
- **Multi-disciplinary** : draws from economics, accounting, statistics, mathematics, operations research.
- **Decision oriented** : concentrates on decision-making and forward planning under uncertainty.
- **Conceptual and metrical** : uses concepts plus measurement (quantitative techniques).

1.3 Scope / Subject matter

Area	Topics covered
Demand analysis	Demand determinants, elasticity, forecasting
Production and cost	Production function, cost, break-even analysis
Pricing	Price-output decisions, pricing policies
Profit management	Profit planning, break-even, risk and uncertainty
Capital management	Capital budgeting, working capital, sources of funds
Market structure	Competition, monopoly, oligopoly

1.4 Significance (Importance)

- Helps in **rational decision making** by choosing the best alternative.
- Gives **tools for optimum use of scarce resources** (men, money, material, machines).
- Helps to **forecast demand, costs and profits** so that the firm can plan ahead.
- Helps in **price fixation, output and capital planning**.
- Helps to **face uncertainty and risk** in the business environment.
- Helps in **co-ordination** between different departments (production, finance, marketing).
- Acts as a base for **business policy formulation** (expansion, diversification, shutdown).

1.5 Functions of Managerial Economics

- **Decision making** : selecting the best course of action among alternatives.
- **Forward planning** : demand, cost and profit forecasts.

- **Resource allocation**: using inputs in the best possible combination.
- **Analysis of business environment**: government policy, competition, trade cycle.
- **Pricing and output decisions** and **capital budgeting**.
- **Controlling**: comparing actual with plans and taking corrective actions.

1.6 Advantages

- Makes managers **analytical** and improves quality of decisions.
- Reduces uncertainty by scientific forecasting.
- Helps in **minimising cost** and **maximising profit**.
- Provides a **rational basis** for investment and expansion.
- Useful for **evaluating government policy effects** on the business.

1.7 Basic economic concepts used by managers

Concept	Meaning
Opportunity cost	Value of the next best alternative given up
Incremental concept	Change in total cost / revenue due to a decision
Time perspective	Short run vs long run effects of a decision
Discounting principle	Rs. 1 today is worth more than Rs. 1 tomorrow
Equi-marginal principle	Allocate resources so that marginal returns are equal in all uses
Marginalism	Decisions are taken at the margin: extra cost vs extra benefit

2. Demand Analysis

2.1 Concept of Demand

- Demand means **desire + ability to pay + willingness to pay** for a commodity at a given price in a given time.
- Mere desire is not demand. Example: a poor man wants a car, but it is not demand.

> Demand = Desire + Purchasing power + Willingness to spend

2.2 Types of Demand

- **Individual demand** and **Market demand** (sum of individual demands).
- **Price demand, income demand, cross demand**.

- **Joint demand** (car and petrol) and **Composite demand** (coal: used for many purposes).
- **Derived demand** (steel from car industry) and **Direct demand** .
- **Durable** and **non-durable goods demand** .

2.3 Demand Function

- Shows the functional relation between quantity demanded and its determinants.

Dx = f(Px, Py, Y, T, A, E, N)

Symbol	Determinant
Px	Price of the commodity itself
Py	Price of related goods (substitutes / complements)
Y	Income of the consumer
T	Tastes, habits and fashion
A / E	Advertisement / expectations about future price
N	Number of consumers (population)

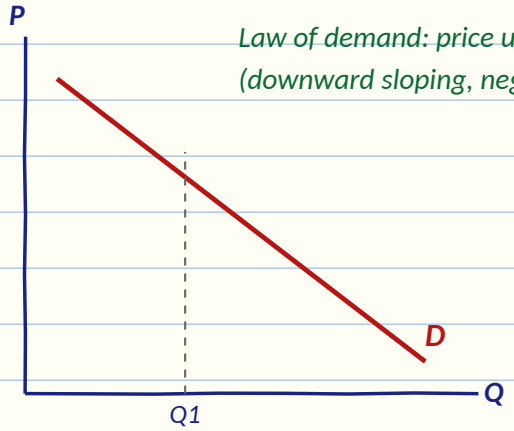
- In a simple form, with other factors constant: **Dx = a - bPx** (linear demand function).

2.4 Law of Demand

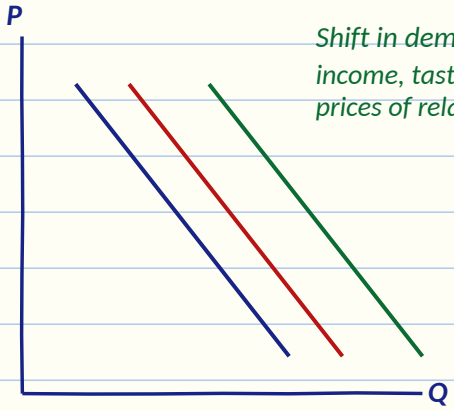
Law: Other things remaining constant, when the price of a commodity rises, the quantity demanded falls, and when price falls, quantity demanded rises. There is an **inverse relationship** between price and quantity demanded. (Alfred Marshall)

Demand schedule

Price (Rs. per unit)	10	8	6	4	2
Quantity demanded	10	20	30	40	50



Law of demand: price up, demand down (downward sloping, negative slope)



Shift in demand: income, tastes, prices of related goods

Assumptions of the law

- Income of the consumer is constant; tastes and preferences do not change.
- Prices of related goods are unchanged; no change in expectations of future price.
- Commodity is **normal** and no new substitutes appear.

Why does the demand curve slope downward?

- **Law of diminishing marginal utility**: the consumer pays less for additional units.
- **Income effect** : fall in price increases real income, so the consumer buys more.
- **Substitution effect** : cheaper good replaces the costlier one.
- **New consumers** enter at lower prices.
- **Different uses** of the commodity (milk, electricity).

Exceptions to the law (upward sloping demand)

- **Giffen goods** (inferior goods, e.g. coarse grain) - Sir Robert Giffen.
- **Veblen / prestige goods** (diamonds, luxury cars): high price attracts buyers.
- **Expectation** of further price rise (shares, gold in a boom).
- **Ignorance**, necessities and emergency purchases (medicines).

Change in quantity demanded vs Change in demand

Basis	Extension / Contraction	Increase / Decrease in demand
Cause	Change in price only	Change in other factors (income, taste)
Movement	Movement along the same curve	Shift of the whole demand curve
Direction	Up or down the curve	Rightward / leftward shift

3. Elasticity of Demand

3.1 Meaning

- Elasticity of demand = **degree of responsiveness** of quantity demanded to a change in its determinants.

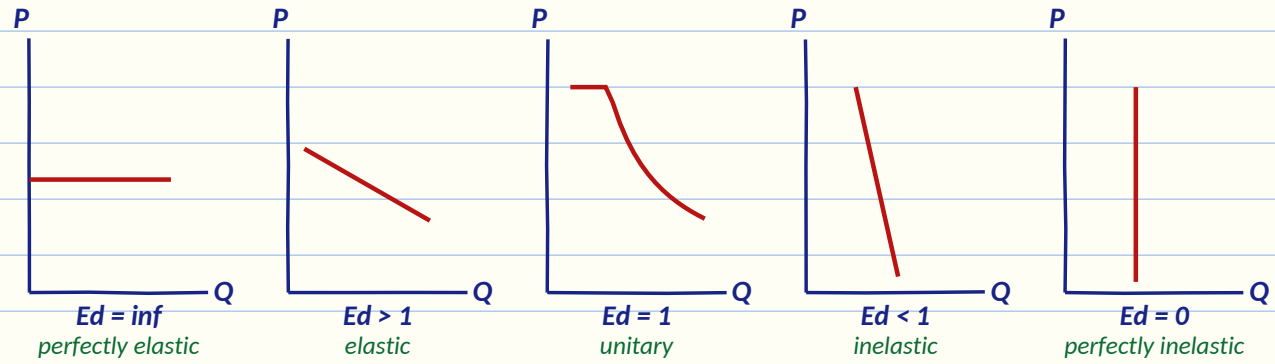
Ed = % change in quantity demanded / % change in price

3.2 Types of elasticity

- **Price elasticity (Ep)** : response of demand to change in price of the good.
- **Income elasticity (Ey)** : response of demand to change in consumer income. Ey = %change in Q / %change in income. Positive for normal goods, negative for inferior goods, greater than 1 for luxuries.
- **Cross elasticity (Exy)** : response of demand for X to change in price of Y. Positive for substitutes (tea and coffee), negative for complements (car and petrol), zero for unrelated goods.
- **Advertisement (promotional) elasticity**: response of sales to change in advertisement expenditure.

3.3 Degrees of Price Elasticity

Type	Value	Meaning
Perfectly elastic	Ed = infinity	Small price change causes infinite change in demand; horizontal curve
Relatively elastic	Ed > 1	% change in demand is more than % change in price (luxuries)
Unitary elastic	Ed = 1	Both change by the same proportion; rectangular hyperbola
Relatively inelastic	Ed < 1	Demand changes less than price (necessities)
Perfectly inelastic	Ed = 0	Demand does not change at all; vertical curve



3.4 Factors affecting price elasticity

- Nature of commodity (necessity: inelastic; luxury: elastic).
- Availability of close **substitutes** (more substitutes: more elastic).
- **Proportion of income** spent on the good (salt: inelastic).
- **Time period** (long run demand is more elastic than short run).
- Number of uses, habits, and price level of the good.

4. Measurement of Elasticity

4.1 Percentage (Proportionate) method - Marshall

Ep = (change in Q / original Q) / (change in P / original P)

Ep = (dQ / dP) x (P / Q)

Example: Price falls from Rs. 10 to Rs. 8 and quantity rises from 100 to 120 units.

%dQ = 20/100 = 20%; %dP = 2/10 = 20% (numerically). **Ep = 20/20 = 1** (unitary elastic).

4.2 Total Outlay (Total Expenditure) method - Marshall

Price	Total expenditure	Elasticity
Falls	Increases	Ed > 1
Falls	Remains same	Ed = 1
Falls	Decreases	Ed < 1

4.3 Point method

Ep = (Lower segment of the demand curve) / (Upper segment)

- On a straight line demand curve: at mid-point Ep = 1; above mid-point Ep > 1; below mid-point Ep < 1; at Y-axis Ep = infinity; at X-axis Ep = 0.

4.4 Arc method

- Used when the change in price is large; uses **average** of the two prices and quantities.

$$Ep = (dQ / dP) \times ((P1 + P2) / (Q1 + Q2))$$

Numerical: P1 = 10, Q1 = 40; P2 = 8, Q2 = 50. dQ = 10, dP = -2.

Ep = (10 / -2) x (18 / 90) = -5 x 0.2 = -1. Numerically **Ep = 1** (unitary).

4.5 Practical importance of elasticity

- Helps in **pricing decisions** (elastic: cut price to raise revenue; inelastic: raise price).
- Used by government for **taxation** (tax inelastic goods like cigarettes, petrol).
- Helps in deciding **wage policy**, devaluation and **international trade** terms.
- Helps a **monopolist** in price discrimination.

5. Demand Forecasting

5.1 Meaning

- Demand forecasting = **scientific estimation of future demand** for a product over a specified period under given conditions.
- Based on past data, present trends and expected future changes.

5.2 Significance

- Planning **production, inventory and manpower**.
- Fixing **sales targets** and preparing the **budget**.
- Controlling cost, deciding **expansion or diversification**, and arranging finance.

5.3 Factors governing demand forecasting

- **Time period** (short term or long term forecast).
- **Level of forecast**: firm level, industry level or economy level.
- **Nature of product**: consumer goods, capital goods, new or established product.
- **General economic conditions**: inflation, income, government policy.
- **Competition** and market structure; **technological changes**.
- **Availability and reliability of data**; cost of forecasting; **purpose** of forecast.
- **Seasonal factors** and consumer behaviour.

5.4 Steps in forecasting

- Identify the objective, then determine the time horizon.
- Select the method, collect and analyse data.
- Make the forecast, then test accuracy and revise.

5.5 Methods of Demand Forecasting



A. Qualitative (Survey) methods

- **Consumer survey**: complete enumeration, sample survey, end-use method; asks buyers about their intentions.
- **Opinion poll of sales force**: salesmen give estimates from their territories (grass-root information).
- **Expert opinion / Delphi method**: panel of experts give anonymous forecasts in rounds until consensus.
- **Test marketing**: new product sold in a small area first, results extended to the whole market.
- **Market simulation / Collective opinion** of executives.

B. Quantitative (Statistical) methods

- **Trend projection**: fitting a line or curve to past data (least squares). **$Y = a + bX$**.
- **Moving average / exponential smoothing**: average of recent periods; gives more weight to recent data.
- **Barometric method**: leading, lagging and coincident indicators of business cycles.
- **Regression analysis**: relation of demand (dependent) with income, price, advertisement (independent variables).
- **Econometric models**: system of equations; **Time series** decomposition (trend, seasonal, cyclical, random).

Simple trend example

Sales for 2020 - 2024: 100, 110, 125, 135, 150. Average yearly rise = (150 - 100)/4 = 12.5.

Forecast for 2025 = 150 + 12.5 = **162.5 units**.

Qualitative vs Quantitative

Basis	Qualitative	Quantitative
Nature	Judgement and opinions	Past data and statistics
Best for	New products, no data	Established products
Cost / time	Less technical	Needs software, skills

6. Managerial Economics and other disciplines

6.1 Relation with Financial Accounting

- Accounting supplies **cost, revenue and profit data** which managerial economics uses for decisions.
- Economics gives the **concepts** (opportunity cost, marginal cost); accounting records **actual historical** data.
- Accounting looks at the past, managerial economics looks at the **future** planning.
- Both aim at profit and efficiency, so they are complementary.

6.2 Relation with Management

- Management = planning, organising, directing, controlling. Managerial economics supplies the **analytical tools** for each function.
- Economics helps in decision making, goal setting and policy; management implements them.

6.3 Other disciplines

Discipline	Contribution
Statistics	Data collection, forecasting, probability
Mathematics	Calculus, optimisation, programming
Operations research	Linear programming, queuing, inventory models
Theory of decision making	Choice under risk and uncertainty
Accounting / Finance	Cost, investment and funding data

Economics vs Managerial Economics: Economics studies all of society's choices (micro + macro, theory). Managerial economics studies the **firm's decisions** (micro, applied, prescriptive).

7. Solved Problems (Elasticity & Forecasting)

Problem 1: Percentage method

The price of a good falls from Rs. 20 to Rs. 15 and quantity demanded rises from 200 to 300 units. Find Ep.

- dQ = 300 - 200 = 100; %dQ = 100/200 x 100 = **50%**

- dP = 15 - 20 = -5; %dP = -5/20 x 100 = **-25%**

Ep = 50 / -25 = -2 (|Ep| = 2 > 1, elastic)

- Total outlay: before = 20 x 200 = 4000; after = 15 x 300 = 4500. Price fell and outlay rose, which confirms **elastic** demand.

Problem 2: Income elasticity

Income rises from Rs. 10,000 to Rs. 12,000 and demand for a good rises from 50 to 60 units. Find Ey.

Ey = (10/50) / (2000/10000) = 0.20 / 0.20 = 1

- Ey = 1, so demand rises in the same proportion as income (unitary income elasticity).

Problem 3: Cross elasticity

Price of coffee rises from Rs. 40 to Rs. 50 and demand for tea rises from 100 to 120 units.

Exy = (20/100) / (10/40) = 0.20 / 0.25 = 0.8

- Exy is positive, so tea and coffee are **substitutes**.

Problem 4: Simple linear trend (least squares)

Sales (in 000 units): 2021 = 20, 2022 = 24, 2023 = 28, 2024 = 32, 2025 = 36. Forecast 2026.

- Take X = -2, -1, 0, 1, 2 for the years. Sum Y = 140, so a = 140/5 = 28.
- Sum XY = (-40) + (-24) + 0 + 32 + 72 = 40; Sum X squared = 10, so b = 40/10 = 4.

Y = 28 + 4X; for 2026, X = 3: Y = 28 + 12 = 40 (000 units)

Problem 5: Moving average

Sales: 10, 12, 14, 16, 18 (3-year moving average).

- First MA = (10+12+14)/3 = 12; second = (12+14+16)/3 = 14; third = (14+16+18)/3 = 16.
- Forecast for next year = **16** (latest moving average).

8. Quick Revision Sheet

Key formulas

Formula	Use
Dx = f(Px, Py, Y, T, A, E, N)	Demand function

Ep = %dQ / %dP	Price elasticity
Ey = %dQ / %dY	Income elasticity
Exy = %dQx / %dPy	Cross elasticity
Arc Ep = (dQ/dP) x (P1+P2)/(Q1+Q2)	Large price change
Y = a + bX	Trend projection

Remember

Ed > 1: price fall raises revenue. Ed = 1: revenue unchanged. Ed < 1: price fall reduces revenue.

Necessities = inelastic; luxuries = elastic; more substitutes = more elastic; long run = more elastic.

Forecasting: Qualitative = survey, Delphi, sales force, test market. Quantitative = trend, regression, barometric, econometric.

One-line differences

- **Demand vs Want**: want has no purchasing power; demand has.
- **Extension vs Increase**: extension is a price effect (movement); increase is a non-price effect (shift).
- **Giffen vs Veblen**: Giffen = inferior necessity; Veblen = luxury status good.

Important Exam Questions

- Define managerial economics. Explain its nature, scope and significance. (10 M)
- State and explain the law of demand. What are its exceptions? (10 M)
- Define elasticity of demand. Explain its types and methods of measurement. (10 M)
- What is demand forecasting? Explain the methods of demand forecasting. (10 M)
- Discuss the relationship of managerial economics with accounting and management. (5 M)
- Short notes: Giffen goods, cross elasticity, Delphi method, opportunity cost.